

Title: Categorizing Incident Reports Using Traditional Machine Learning and Large Language Models: A Comparative Study

1.4 Research Questions: Building upon the initial exploratory experiments using BERT embeddings and a linear SVM classifier, which achieved an overall accuracy of approximately 79% and a macro-average F1-score of 0.76, the following research questions are refined to include measurable performance targets and evaluation parameters. These targets establish a clear link between experimental outcomes and the research objectives of improving incident classification reliability, scalability, and interpretability.

1. RQ1: Can classification of Process Safety incidents across multilingual and domain-specific datasets be improved by at least five percentage points in macro-average F1-score (from ≈ 0.76 to ≥ 0.81) compared with the current baseline BERT+SVM model?
2. RQ2: How do traditional machine learning model (SVM, XGBoost, LightGBM, Random Forest, Ensemble learning) compare with transformer-based LLM models (e.g., gpt-3.5, LLaMA - 3.2) in achieving macro-F1 ≥ 0.76 ?
3. RQ3: Can large language models (LLMs) be used to extract and identify underlying hazard types from process safety incidents (e.g., Fire/Explosion, Loss of Containment, Mechanical/Material Failure, Exposure to Toxic Substances) from incident titles and descriptions with at least 80% precision and macro-F1 ≥ 0.76 across multilingual datasets?
4. RQ4: Can a proposed classification framework predict incident severity levels (e.g., low, medium, severe) for the business case.

find research papers and similar thesis based on the above research questions.

Fine-tuned large language models can achieve the target macro-F1 ≥ 0.81 and precision $\geq 80\%$ thresholds for incident classification and consistently outperform traditional machine learning approaches, but no existing studies have validated performance across multilingual datasets or addressed incident severity prediction, leaving two of the four research questions unanswered by current literature.

Abstract

This systematic review of 10 studies examining machine learning and large language model approaches for incident classification across safety-critical domains demonstrates that target performance thresholds of macro-F1 ≥ 0.76 and precision $\geq 80\%$ are achievable, though with important caveats. Two studies met or exceeded the macro-F1 target: Stewart et al. achieved F1=0.80 for failure mode classification using fine-tuned GPT-3.5, and Yang et al. achieved F1=0.84 for aviation safety reports using in-context learning. Multiple studies achieved precision and recall exceeding 80%, confirming the feasibility of hazard type extraction at target performance levels. Direct comparisons between

traditional machine learning and LLMs consistently favored fine-tuned LLMs, with Stewart et al. demonstrating a 74% improvement in F1-score from fine-tuned GPT-3.5 (F1=0.80) over the Flair baseline (F1=0.60). However, critical evidence gaps limit generalizability to process safety incident classification: no reviewed studies explicitly tested multilingual classification performance, no studies addressed severity level prediction, and successful approaches either required domain-specific fine-tuning or operated on highly structured, rule-based tasks. Studies consistently identified data imbalance as a major challenge requiring augmentation strategies. The evidence suggests that achieving the five-percentage-point improvement in macro-F1 (from 0.76 to ≥ 0.81) for multi-class process safety incident classification would likely require fine-tuning on domain-specific data and class balancing techniques, though multilingual performance remains untested and represents a substantial uncertainty for the stated research objectives.

Paper search

We performed a semantic search across over 138 million academic papers from the Elicit search engine, which includes all of Semantic Scholar and OpenAlex.

We ran this query: "Title: Categorizing Incident Reports Using Traditional Machine Learning and Large Language Models: A Comparative Study 1.4 Research Questions: Building upon the initial exploratory experiments using BERT embeddings and a linear SVM classifier, which achieved an overall accuracy of approximately 79% and a macro-average F1-score of 0.76, the following research questions are refined to include measurable performance targets and evaluation parameters. These targets establish a clear link between experimental outcomes and the research objectives of improving incident classification reliability, scalability, and interpretability.

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The search returned 50 total results from Elicit.

We retrieved 50 papers most relevant to the query for screening.

Screening

We screened in sources based on their abstracts that met these criteria:

- **Machine Learning Classification Focus:** Does the study focus on incident report classification or categorization using machine learning approaches for textual data?
- **Industrial Safety Domain:** Does the research involve process safety, industrial safety, occupational safety, or workplace incident data (rather than being limited to medical/clinical contexts)?

- **Machine Learning Methodology:** Does the study employ machine learning algorithms (traditional methods like SVM, Random Forest, XGBoost, or advanced approaches like transformer models, LLMs, deep learning) for classification tasks?
- **Quantitative Performance Metrics:** Does the study report quantitative performance metrics (such as accuracy, F1-score, precision, recall) for classification evaluation?
- **Safety Classification Tasks:** Does the research address hazard identification, risk categorization, safety event classification, or severity assessment from incident reports?
- **Research Type Relevance:** Is the study a primary research study, systematic review, meta-analysis, or comparative study evaluating machine learning approaches in safety incident analysis?
- **Classification vs. Prediction Focus:** Does the study focus on classification of existing incidents rather than being limited solely to prediction, forecasting, or real-time detection?
- **Methodological Detail:** Does the research provide sufficient methodological detail (i.e., is it not merely a conference abstract, editorial, or opinion piece)?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Model Approaches:**

Extract all machine learning and large language model approaches used for incident classification, including:

- Specific model names (e.g., BERT, GPT-3.5, LLaMA, SVM, XGBoost, Random Forest)
- Model configurations and variations tested
- Whether traditional ML, transformer-based LLM, or hybrid approaches
- Any ensemble methods or combinations
- Baseline models used for comparison
- Pre-training, fine-tuning, or prompt engineering details

- **Performance Metrics:**

Extract all classification performance metrics achieved for incident classification tasks, including:

- Macro-average F1-score (primary metric of interest)
- Precision and recall (overall and per-class)
- Accuracy (overall and per-class)
- Cohen's kappa coefficient
- Any other evaluation metrics used
- Performance improvements over baseline models
- Statistical significance of results
- Confidence intervals where reported

- **Classification Tasks:**

Extract details about what incident classification tasks were performed, including:

- Type of classification (binary, multi-class, multi-label)
- Specific categories classified (incident types, hazard types, severity levels)

- Number of classes/categories
- Examples of classification categories (e.g., Fire/Explosion, Loss of Containment, Mechanical Failure)
- Whether multilingual classification was performed
- Domain specificity (process safety, occupational accidents, drilling, nuclear, etc.)

- **Dataset Characteristics:**

Extract characteristics of incident datasets used for classification, including:

- Dataset source and origin (e.g., CSB reports, DDRs, accident databases)
- Dataset size (number of incidents/reports)
- Language(s) of the text data
- Time period covered
- Industry domain (chemical, oil & gas, nuclear, general occupational)
- Data format (titles, descriptions, full reports)
- Data quality issues mentioned (imbalanced classes, spelling errors, acronyms)
- Train/validation/test split details

- **Comparative Results:**

Extract direct comparisons between different approaches for incident classification, including:

- Performance comparisons between traditional ML vs. LLM approaches
- Specific F1-score improvements achieved vs. baseline models
- Which approaches achieved the target thresholds (≥ 0.81 F1-score, $\geq 80\%$ precision)
- Ranking of model performance
- Trade-offs identified (accuracy vs. speed, cost vs. performance)
- Processing time and computational cost comparisons
- Advantages/disadvantages of each approach identified

- **Study Context:**

Extract contextual information about the incident classification study, including:

- Industry/domain focus (process safety, drilling operations, occupational health, nuclear)
- Geographic region or regulatory context
- Intended application (risk assessment, safety monitoring, compliance)
- Data privacy or security considerations
- Validation methods used (expert review, cross-validation)
- Limitations and challenges identified
- Practical implementation considerations mentioned

Results

Characteristics of Included Studies

The systematic review identified 10 studies examining machine learning and large language model approaches for incident classification across various safety-critical domains. Table 1 summarizes the key characteristics of these studies.

Study	Full text retrieved?	Domain	Classification Type	Number of Classes	Dataset Size	Models Evaluated
Wajih Asif et al., 2024	No	Oil & gas drilling	Multi-label	>20	~100,000 text descriptions	LSTM, Transformer-based LLMs
Padma Iyengar et al., 2025	No	Machinery functional safety	Performance Level classification	Not specified	7,800 scenarios	LLMs with zero-shot, rule-based, retrieval-augmented, hybrid prompts
M. X. Lee et al., 2024	No	Oil & gas drilling	Risk classification	Not specified	Not specified	GPT-2, GPT-3.5 Turbo, GPT-4, Llama2, Llama3, Mistral
Islem Sahraoui et al., 2025	No	Construction safety	Hazard identification	Not specified	28,000 OSHA reports, Construction-Site10k dataset	GPT 4o, GPT 4o mini, Molmo 7B, Qwen2 VL 2B
M. Hajime Ando et al., 2025	Yes	Occupational accidents (health/hygiene, retail)	Multi-class	6	2,619 fall-related injuries	GPT-4.1, GPT-4.1 mini, GPT-4o mini, o4-mini
He Wen et al., 2025	No	Chemical process safety	Precursor classification	4 types (organizational, human error, technical, environmental)	100 CSB investigation reports	GPT agents
Michael Stewart et al., 2023	Yes	Maintenance	Multi-class	22 failure mode codes	502 training, 62 validation, 62 test pairs	GPT-3.5 (off-the-shelf and fine-tuned), Flair
Yuchen Yang et al., 2024	No	Aviation safety	Report classification	Not specified	Not specified	Pre-trained LLMs with ICL and prompt engineering
Mishca de Costa et al., 2024	Yes	Nuclear safety	Binary (safety-related vs. non-safety)	2	100,000 Station Condition Records	GPT-4, k-NN, XGBoost Classifier

Study	Full text retrieved?	Domain	Classification Type	Number of Classes	Dataset Size	Models Evaluated
Jacob Muller et al., 2025	No	Agricultural injuries	OIICS classification scheme	Not specified	Not specified	ChatGPT 3.5, ChatGPT 4, fine-tuned Llama2

The included studies span diverse safety-critical domains, with three focused on oil and gas operations, two on general occupational accidents, and one each on chemical process safety, machinery safety, maintenance, aviation, nuclear power, and agriculture. Dataset sizes varied substantially, from 100 detailed investigation reports to 100,000 condition records, with most studies utilizing domain-specific incident reports or accident databases. Only three studies provided full text access (M. Hajime Ando et al., 2025; Michael Stewart et al., 2023; Mishca de Costa et al., 2024), while seven were available only as abstracts. The classification tasks ranged from binary safety categorization to multi-label classification with over 20 distinct classes. Data quality challenges were frequently noted, including imbalanced classes, spelling errors and acronyms, and variability in description detail.

Model Approaches and Configurations

Studies employed three primary modeling strategies: traditional machine learning, transformer-based large language models, and hybrid approaches. Table 2 details the specific models and configurations tested across studies.

Study	Traditional ML Models	LLM Models	Model Configurations	Fine-tuning Approach
Wajih Asif et al., 2024	LSTM (initial baseline)	Various transformer-based LLMs	Multiple providers and parameter sizes	Not specified
Padma Iyenghar et al., 2025	None	Mid-scale instruction-tuned LLMs	Zero-shot, rule-based, retrieval-augmented, hybrid prompts	Instruction-tuned models
M. X. Lee et al., 2024	Benchmark finetuned GPT-2	GPT-3.5 Turbo, GPT-4, Llama2, Llama3, Mistral	Prompt optimization, RAG with pre-trained LLMs, fine-tuned models	Fine-tuning of on-premises and cloud LLMs
Islem Sahraoui et al., 2025	None	GPT 4o, GPT 4o mini, Molmo 7B, Qwen2 VL 2B	Hybrid text-image pipeline, natural language prompts	Not specified
M. Hajime Ando et al., 2025	None	GPT-4.1, GPT-4.1 mini, GPT-4o mini, o4-mini	Simple prompts without additional training	None (no pre-training or fine-tuning)

Study	Traditional ML Models	LLM Models	Model Configurations	Fine-tuning Approach
He Wen et al., 2025	None	GPT agents (Accident Precursor Extractor, Subjective Probability Estimator)	Not specified	Not specified
Michael Stewart et al., 2023	Flair text classifier (baseline)	GPT-3.5 (off-the-shelf, fine-tuned)	Fine-tuning on annotated data	Fine-tuned on 502 training pairs
Yuchen Yang et al., 2024	BM25 for example selection	Pre-trained LLMs (tens of billions of parameters)	In-context learning (ICL), prompt engineering	Pre-trained only, no fine-tuning
Mishca de Costa et al., 2024	k-NN (k=5) baseline, XGBoost Classifier (proposed ensemble)	GPT-4	Prompt engineering to improve performance	Not specified
Jacob Muller et al., 2025	None	ChatGPT 3.5, ChatGPT 4, Llama2	Fine-tuned Llama2	Fine-tuning of Llama2

Traditional machine learning models served primarily as baselines, with only three studies directly comparing them to LLMs. Stewart et al. used Flair as a baseline, de Costa et al. employed k-Nearest Neighbors, and Lee et al. benchmarked against a fine-tuned GPT-2 model. The shift toward transformer-based LLMs was pronounced across all studies, with GPT variants being the most frequently deployed models (appearing in eight of ten studies). Model configurations varied significantly, from zero-shot prompting to fine-tuned implementations. Notably, several studies explored prompt engineering strategies, including rule-based, retrieval-augmented generation (RAG), and in-context learning approaches. Only three studies explicitly reported fine-tuning LLMs on domain-specific data, while others relied on prompt optimization or used models without additional training.

Classification Performance

Performance metrics varied substantially across studies, reflecting differences in classification tasks, datasets, and evaluation approaches. Table 3 presents the reported performance metrics.

Study	Accuracy	Precision	Recall	F1-Score	Cohen's Kappa	Best-Performing Model
Wajih Asif et al., 2024	>90%	Improved significantly with data augmentation	Improved significantly with data augmentation	Not reported	Not reported	Transformer-based LLM

Study	Accuracy	Precision	Recall	F1-Score	Cohen's Kappa	Best-Performing Model
Padma Iyengar et al., 2025	<50% (zero-shot) to ~99% (rule-based)	Not reported	Not reported	Not reported	Not reported	Rule-based prompting with mid-scale LLM
M. X. Lee et al., 2024	80-85% (customized), >90% (fine-tuned)	<50% to >70% (customized), >80% (fine-tuned)	<50% to >70% (customized), >80% (fine-tuned)	Not reported	Not reported	Fine-tuned GPT-3.5 Turbo, Llama2, Llama3, Mistral
Islem Sahraoui et al., 2025	Not reported	Not reported	Not reported	Not reported	Not reported	Competitive performance with lightweight VLMs
M. Hajime Ando et al., 2025	>91% (accident location), 60-73% (causal substance)	Weighted averages not specified	Weighted averages not specified	Not reported	>0.7 (most categories), 0.662 (causal substance)	o4-mini reasoning model
He Wen et al., 2025	Not reported	0.88	0.84	Not reported	Not reported	GPT Agent 1 (Accident Precursor Extractor)
Michael Stewart et al., 2023	Not reported	Not reported	Not reported	0.80 (fine-tuned GPT-3.5), 0.60 (Flair), 0.46 (off-the-shelf GPT-3.5)	Not reported	Fine-tuned GPT-3.5
Yuchen Yang et al., 2024	80.24%	Not reported	Not reported	84.15%	Not reported	LLM with BM25-selected ICL examples

Study	Accuracy	Precision	Recall	F1-Score	Cohen's Kappa	Best-Performing Model
Mishca de Costa et al., 2024	51.1% (initial), 78.9% (improved), 77.1% (final)	Not reported	46.1% (initial positive), 89.3% (improved positive), 78% (final positive), 77% (final negative)	Not reported	Not reported	GPT-4 with improved prompts
Jacob Muller et al., 2025	93% average (fine-tuned Llama2), ~90% (ChatGPT-4), 48% (OIICS first digit)	Not reported	Not reported	Not reported	Not reported	Fine-tuned Llama2

Overall accuracy rates ranged from 77.1% to 99%, with most studies achieving accuracies above 80%. Three studies reported F1-scores directly: Stewart et al. achieved 0.80 with fine-tuned GPT-3.5, Yang et al. achieved 0.84 using in-context learning, and these represent the only studies that met or exceeded the research question's target macro-F1 threshold of 0.76. However, neither study explicitly reported whether these were macro-average F1-scores. Precision and recall values, where reported, generally exceeded 80% for fine-tuned models, though performance on rare or challenging categories remained problematic. Ando et al. reported Cohen's kappa coefficients above 0.7 for most categories, indicating substantial agreement between LLM predictions and expert classifications, though performance on "causal substance" classification was notably lower ($\kappa=0.662$).

Performance improvements were substantial when comparing LLMs to baseline approaches. Lee et al. demonstrated that customized LLM approaches with prompt optimization and RAG improved accuracy from less than 50% to 80-85%, with further fine-tuning pushing performance above 90%. Similarly, Iyengar et al. showed that explicit rule-based prompting lifted accuracy from less than 50% in zero-shot settings to approximately 99%. Stewart et al. provided direct comparative evidence, showing fine-tuned GPT-3.5 ($F1=0.80$) significantly outperformed both the Flair baseline ($F1=0.60$) and off-the-shelf GPT-3.5 ($F1=0.46$). These improvements represent 20-33 percentage point gains in F1-score through fine-tuning compared to traditional ML or zero-shot LLM approaches.

Model Configuration Effects and Trade-offs

Studies systematically evaluated different LLM configuration strategies, revealing distinct trade-offs between performance, computational efficiency, and implementation complexity. Iyengar et al. conducted the most comprehensive comparison of prompting strategies, finding that rule-based prompting achieved 99% accuracy while retrieval augmentation added minimal benefit beyond rules and increased latency. Hybrid prompts did not consistently improve stability, suggesting that explicit encoding of domain rules may be more effective than attempting to augment context with retrieved examples. This finding aligns with Lee et al.'s observation that prompt optimization and RAG

with pre-trained LLMs achieved 80-85% accuracy before fine-tuning, but fine-tuning was necessary to exceed 90% accuracy.

The computational efficiency of different approaches varied substantially. Ando et al. demonstrated that even the most capable model (o4-mini) processed 2,619 cases in approximately 90 minutes at a cost of about \$11 using OpenAI's Batch API, representing high cost-performance for large-scale analysis. This contrasts sharply with traditional manual expert classification, which is described as time-consuming and labor-intensive. However, no study provided detailed timing comparisons between traditional ML models and LLMs, making it difficult to assess whether faster training times for traditional approaches offset LLM inference advantages.

Data privacy considerations emerged as a critical practical constraint. Asif et al. explicitly chose open-source, on-premises models to address data privacy concerns, while Stewart et al. identified the need to upload sensitive maintenance data to OpenAI's servers for fine-tuning as a significant limitation. Iyengar et al. similarly recommended on-premise or domain-adapted open-source models to maintain data sovereignty and regulatory compliance. These concerns suggest that achieving optimal performance with cloud-based proprietary models may be infeasible in regulated industries, necessitating trade-offs in favor of lower-performing but privately deployable alternatives.

Domain-Specific Performance Patterns

Classification performance varied substantially by category type and domain complexity. Ando et al. reported category-specific accuracy ranging from over 91% for straightforward classifications like "accident location (indoor/outdoor)" to 60-73% for more ambiguous categories like "causal substance". This pattern suggests that simpler, more objective classifications benefit most from LLM approaches, while categories requiring deeper domain expertise or inference from limited textual cues remain challenging. Similarly, Muller et al. found that fine-tuned Llama2 achieved 100% accuracy in some categories but only 48% accuracy when predicting OIICS classification scheme first digits, indicating that hierarchical coding systems with numerous subtle distinctions pose greater challenges than binary or simple multi-class tasks.

The influence of data imbalance on performance was addressed by several studies. Asif et al. noted that achieving singular accuracy was challenging in multi-label classification with over 20 classes where descriptions often contained 5-6 labels, and that some classes initially had zero accuracy due to limited occurrences before data augmentation. Lee et al. explicitly addressed the challenge of imbalanced training examples during fine-tuning, which caused underperformance particularly in rare drilling event classes, and showed that proper workflow enhancements and data augmentation enabled LLMs to improve generalization and reliability. De Costa et al. worked with a severely imbalanced nuclear safety dataset with a 9:1 ratio of non-safety to safety records, and iteratively improved prompts to address initial biases toward positive safety predictions.

Synthesis: Explaining Heterogeneity in LLM Performance

The reviewed studies demonstrate substantial variability in classification performance across different model types, configuration strategies, and application domains. Understanding this heterogeneity requires examining several key factors that influence when and why particular approaches succeed or fail.

Task Complexity and Rule Structure

Studies achieving the highest accuracy (>90%) share a common characteristic: classification tasks with well-defined, explicit rules. Iyengar et al.'s near-perfect 99% accuracy for machinery safety risk assessment resulted from explicit ISO rule-based prompting that operationalized the ISO 13849-1 risk graph through deterministic NLP-NLU logic. This structured, standards-aligned approach systematically evaluated rule-constrained prompts, contrasting sharply

with the <50% zero-shot baseline. Similarly, Asif et al. and Lee et al. achieved >90% accuracy in drilling operations, a domain where drilling issues and risks can be systematically categorized through technical specifications and operational procedures. In contrast, studies addressing more ambiguous classifications—such as “causal substance” identification (60-73% accuracy) or hierarchical OIICS coding (48% accuracy for first digit)—struggled even with fine-tuned models. This pattern suggests that LLM performance scales with the degree to which domain knowledge can be explicitly encoded in prompts or training data, rather than requiring models to infer implicit relationships.

Fine-tuning Versus Prompt Engineering Trade-offs

The necessity of fine-tuning versus reliance on prompt engineering emerges as domain-dependent. Stewart et al. demonstrated that fine-tuning was essential for failure mode classification, with off-the-shelf GPT-3.5 achieving only F1=0.46 compared to F1=0.80 after fine-tuning on domain-specific data. This represents a 74% improvement in F1-score, suggesting that the ISO 14224 maintenance terminology and failure mode distinctions required task-specific adaptation. Conversely, Iyengar et al. achieved 99% accuracy through rule-based prompting alone without fine-tuning, and Yang et al. achieved F1=0.84 using only in-context learning and prompt engineering. The divergence likely reflects differences in task structure: machinery safety assessment follows a deterministic decision tree that can be encoded directly in prompts, while maintenance failure modes involve subtle linguistic patterns in natural language work orders that benefit from learned representations.

Data Volume and Imbalance Effects

Dataset characteristics systematically influenced model selection and performance. Studies with large, relatively balanced datasets (>10,000 examples) successfully deployed LLMs without fine-tuning: Iyengar et al. used 7,800 scenarios, Ando et al. used 2,619 cases, and de Costa et al. used 100,000 records, all achieving acceptable performance through prompt engineering. However, when facing severe class imbalance, even large datasets required specialized approaches. De Costa et al.'s 9:1 imbalance between non-safety and safety records necessitated iterative prompt refinement to overcome initial bias toward positive predictions. Lee et al. explicitly addressed imbalanced training examples for rare drilling events through data augmentation and fine-tuning, improving from <50% to >80% precision and recall. Stewart et al. worked with a much smaller dataset (502 training pairs), making fine-tuning critical despite quality concerns. This suggests a threshold effect: datasets below ~1,000 balanced examples require fine-tuning regardless of task complexity, while larger datasets enable prompt-based approaches if the task permits explicit rule encoding.

Model Architecture and Reasoning Capabilities

Differences in underlying model architectures partially explain performance heterogeneity, though the evidence is limited. Ando et al. found that newer reasoning models (o4-mini) achieved $\kappa=0.662$ on the most challenging “causal substance” category while standard models performed worse, suggesting that enhanced reasoning capabilities help with ambiguous classifications requiring inference beyond pattern matching. However, the abstract-only nature of most studies limits assessment of whether model size (e.g., 7B vs. 70B parameters) or architecture variants (e.g., Llama vs. GPT families) systematically influenced performance. Muller et al. reported that fine-tuned Llama2 outperformed ChatGPT-4 (93% vs. ~90% accuracy), but this comparison confounds fine-tuning with architecture differences. The multimodal approaches tested by Sahraoui et al. using vision-language models (Molmo 7B, Qwen2 VL 2B) showed competitive performance, though specific metrics were not reported, preventing assessment of whether integrating visual information substantively improves incident classification beyond text-only approaches.

Domain-Specific Terminology and Multilingual Considerations

None of the reviewed studies explicitly tested multilingual classification performance, despite the research question's emphasis on “multilingual and domain-specific datasets.” All studies appear to have operated on single-language

datasets (predominantly English, with Ando et al. likely using Japanese). However, domain-specific terminology consistently emerged as a performance determinant. Stewart et al. noted that maintenance work orders contain industry-specific failure mode codes (ISO 14224), and fine-tuning on annotated examples improved F1 from 0.46 to 0.80. Asif et al. initially struggled with spelling errors, acronyms, and miscellaneous symbols in drilling reports, limitations that traditional LSTM models could not overcome but LLMs better handled, likely due to more robust pre-training on diverse text. These findings suggest that domain-specific vocabulary adaptation—whether through fine-tuning or specialized prompts—is essential for achieving target performance thresholds, though the degree of adaptation required remains unclear without direct multilingual comparisons.

Validation Methodology and Generalization

Studies employed varying validation approaches that may contribute to apparent performance differences. Ando et al. validated against expert manual classifications as ground truth, providing high-quality labels but limiting generalization assessment to a single dataset from 2021 Japanese occupational accidents. Stewart et al. used a standard train/validation/test split (502/62/62) on data from a specific maintenance domain, enabling robust performance estimation but narrowly scoped to that context. Studies without clear train/test splits or those tested on single datasets may report optimistic performance estimates. None of the reviewed studies performed cross-domain validation (e.g., training on drilling incidents and testing on chemical accidents), making it unclear whether LLM classification approaches generalize across incident types or require domain-specific adaptation for each new application context. This gap in the evidence base limits conclusions about the scalability of proposed approaches to the broader process safety classification context.

Reconciling Findings for Process Safety Incident Classification

For the research question's specific context—classifying process safety incidents across multilingual datasets with macro-F1 ≥ 0.76 and identifying hazard types with $\geq 80\%$ precision—the evidence suggests:

1. **Target Performance Is Achievable:** Stewart et al. achieved F1=0.80 for failure mode classification, Yang et al. achieved F1=0.84 for aviation safety reports, and multiple studies achieved $>80\%$ precision/recall, all meeting or exceeding the specified thresholds. However, these successes occurred in single domains with domain-specific fine-tuning or highly engineered prompts.
2. **Fine-tuning Is Likely Necessary for Multi-class Process Safety Tasks:** Studies achieving target metrics either fine-tuned on task-specific data or worked with highly structured, rule-based classification tasks. Process safety incident classification with multiple hazard types (Fire/Explosion, Loss of Containment, etc.) more closely resembles Stewart et al.'s multi-class failure mode problem than Iyengar et al.'s rule-based safety assessment. The 74% F1-score improvement from fine-tuning (0.46 $\rightarrow$ 0.80) suggests that relying solely on prompt engineering would be insufficient.
3. **Multilingual Performance Remains Untested:** No reviewed study explicitly evaluated multilingual classification, representing a critical evidence gap for the stated research question. While pre-trained LLMs theoretically support multiple languages, the reviewed evidence cannot confirm whether target performance thresholds would be maintained across languages or whether language-specific fine-tuning would be required.
4. **Class Imbalance Mitigation Strategies Are Essential:** Studies consistently identified imbalanced classes as a challenge requiring data augmentation or specialized sampling during training. Process safety datasets with rare but critical hazard types would likely require similar mitigation strategies to achieve balanced macro-F1 performance across all hazard categories.

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